

Ex1

$$C^1 = \begin{pmatrix} 5 & 1 & 4 & 7 \\ 6 & 2 & 2 & 6 \\ 2 & 8 & 4 & 4 \\ 3 & 5 & 7 & 1 \end{pmatrix}$$

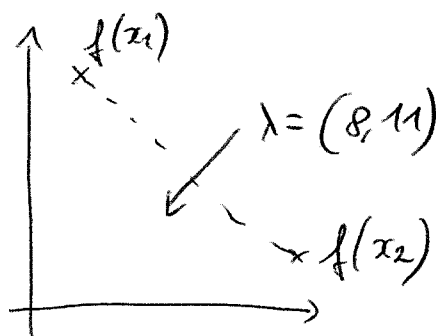
$$C^2 = \begin{pmatrix} 3 & 6 & 4 & 2 \\ 1 & 3 & 8 & 3 \\ 5 & 2 & 2 & 3 \\ 4 & 2 & 3 & 5 \end{pmatrix}$$

$$\lambda = (1, 0) : 1C^1 + 0.C^2 = \begin{pmatrix} 5 & \boxed{1} & 4 & 7 \\ 6 & 2 & \boxed{2} & 6 \\ \boxed{2} & 8 & 4 & 4 \\ 3 & 5 & 7 & \boxed{1} \end{pmatrix} \mapsto f(x_1) = (6, 24)$$

$$\lambda = (0, 1) : 0C^1 + 1.C^2 = \begin{pmatrix} 3 & 6 & 4 & \boxed{2} \\ \boxed{1} & 3 & 8 & 3 \\ 5 & \boxed{2} & 2 & 3 \\ 4 & 2 & \boxed{3} & 5 \end{pmatrix} \mapsto f(x_2) = (28, 8)$$

entre $f(x_1)$ et $f(x_2)$:

$$\begin{aligned} \lambda &= (f_2(x_1) - f_2(x_2), f_1(x_2) - f_1(x_1)) \\ &= (24 - 8, 28 - 6) = (16, 22) = (8, 11) \end{aligned}$$



$$\lambda_1 C^1 = \begin{pmatrix} 40 & 8 & 32 & 56 \\ 48 & 16 & 16 & 48 \\ 16 & 64 & 32 & 32 \\ 24 & 40 & 56 & 8 \end{pmatrix}$$

$$\lambda_2 C^2 = \begin{pmatrix} 33 & 66 & 44 & 22 \\ 11 & 33 & 88 & 33 \\ 55 & 22 & 22 & 33 \\ 44 & 22 & 33 & 55 \end{pmatrix}$$

$$\begin{pmatrix} \boxed{73} & 74 & 76 & 78 \\ 59 & \boxed{49} & 104 & 81 \\ 71 & 86 & \boxed{54} & 65 \\ 68 & 62 & 89 & \boxed{63} \end{pmatrix} \mapsto f(x_3) = (12, 13)$$

entre $f(x_1)$ et $f(x_3)$:

$$\begin{aligned} \lambda &= (f_2(x_1) - f_2(x_3), f_1(x_3) - f_1(x_1)) \\ &= (24 - 13, 12 - 6) = (11, 6) \end{aligned}$$

$$\lambda_1 C^1 = \begin{pmatrix} 55 & 11 & 44 & 77 \\ 66 & 22 & 22 & 66 \\ 22 & 88 & 44 & 44 \\ 33 & 55 & 77 & 11 \end{pmatrix} \quad \lambda_2 C^2 = \begin{pmatrix} 18 & 36 & 24 & 12 \\ 6 & 18 & 48 & 18 \\ 30 & 12 & 12 & 18 \\ 24 & 12 & 18 & 30 \end{pmatrix}$$

$$\begin{pmatrix} 73 & \boxed{47} & 68 & 89 \\ 72 & 40 & \boxed{70} & 84 \\ \boxed{52} & 100 & 56 & 62 \\ 57 & 67 & 95 & \boxed{41} \end{pmatrix} \rightarrow \text{m solution que } x_1 \rightarrow (6, 24)$$

On s'arrête entre $f(x_1)$ et $f(x_3)$.

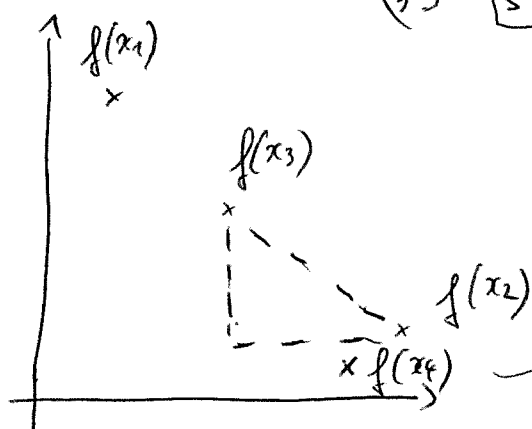
Entre $f(x_3)$ et $f(x_2)$:

$$\lambda = (f_2(x_3) - f_2(x_2), f_1(x_2) - f_1(x_3)) \\ = (13 - 8, 28 - 12) = (5, 16)$$

$$\lambda_1 C^1 = \begin{pmatrix} 25 & 5 & 20 & 35 \\ 30 & 10 & 10 & 30 \\ 10 & 40 & 20 & 20 \\ 15 & 25 & 35 & 5 \end{pmatrix}$$

$$\lambda_2 C^2 = \begin{pmatrix} 48 & 86 & 64 & 32 \\ 16 & 48 & 128 & 48 \\ 80 & 32 & 32 & 48 \\ 64 & 32 & 48 & 80 \end{pmatrix}$$

$$\begin{pmatrix} 73 & 103 & 84 & \textcircled{67} \\ \textcircled{46} & 58 & 138 & 78 \\ 90 & 72 & \textcircled{52} & 68 \\ 79 & \textcircled{57} & 83 & 85 \end{pmatrix} \rightarrow f(x_4) = (22, 7)$$



Solution au dehors du triangle
formé par $f(x_3)$ et $f(x_2)$ mais
qui domine $f(x_2)$.
→ on élimine $f(x_2)$ et on
recommence entre $f(x_3)$ et $f(x_4)$

Entre $f(x_3)$ et $f(x_4)$:

$$\lambda = (f_2(x_3) - f_2(x_4), f_1(x_4) - f_1(x_3)) \\ = (13 - 7, 22 - 12) = (6, 10) = (3, 5)$$

$$\lambda_1 C^1 = \begin{pmatrix} 15 & 3 & 12 & 21 \\ 18 & 6 & 6 & 18 \\ 6 & 24 & 12 & 12 \\ 9 & 15 & 21 & 3 \end{pmatrix}$$

$$\lambda_2 C^2 = \begin{pmatrix} 15 & 30 & 20 & 10 \\ 5 & 15 & 40 & 15 \\ 25 & 10 & 10 & 15 \\ 20 & 10 & 15 & 25 \end{pmatrix}$$

$$\begin{pmatrix} \textcircled{30} & 33 & 32 & 31 \\ 23 & \textcircled{21} & 46 & 37 \\ 31 & 34 & \textcircled{22} & 27 \\ 28 & 25 & 36 & \textcircled{28} \end{pmatrix} \rightarrow \text{on retire la solution } x_3. \\ \rightarrow \text{on s'arrête.}$$

On trouve donc $\tilde{Y}_{SE} = \{(6, 24), (12, 13), (22, 7)\}$